

Pathology foundation-model research-software demonstration

29 September 2026

Contents

Research-software output: internally derived TCGA estimates	1
COAD - TCGA-AA-A01F - TITAN	1
Prominent TCGA tissue-source-site-code sensitivity warning	1
Case context (not used as model input)	2
Continuous internally derived estimates	4
Binary research-model calls	5
Binary development reliability	6
Quality and provenance	6
Endpoint interpretation and target sources	6

Research-software output: internally derived TCGA estimates

COAD - TCGA-AA-A01F - TITAN

UNVALIDATED RESEARCH OUTPUT - REFERENCE RANK, NOT PROBABILITY - NO INDEPENDENT EXTERNAL VALIDATION - NOT FOR DIAGNOSIS, TREATMENT SELECTION OR CLINICAL RISK ESTIMATION

Models applied	Continuous	Binary	Slides pooled
23	15	8	1

This research-software demonstration applies models developed and evaluated only in TCGA. Every displayed value is an **internally derived TCGA estimate**, not external performance evidence or a diagnostic output. Binary LDA scores are uncalibrated. Every displayed **Reference rank, not probability** describes relative position only.

For every cancer and representation, PathoFMPred displays only models that met the recorded representation-specific predictability rule. In the matched three-representation atlas, this means Q2 at least 0.20 for a continuous endpoint or AUROC at least 0.60 for a binary endpoint. TITAN tasks outside the common cohort must pass the supporting permutation/FDR screen. By default, the report also excludes continuous models developed with fewer than 100 labelled participants and binary models with fewer than 50 participants in either class. Explicit opt-in can add only these limited-sample threshold-crossing models. Tested-below-threshold reference objects never enter prediction reports.

Prominent TCGA tissue-source-site-code sensitivity warning

13 model(s) in this output fell below their original documented screening threshold under internal validation grouped by TCGA tissue-source-site code. These calls must not be highlighted without this warning. The barcode-derived code is not an institution, scanner or laboratory identifier, and this grouping is not external validation.

Endpoint (type)	TSS-grouped metric	Grouped minus random	Warning category
APC (binary)	BA 0.519	-0.258	Near chance or worse
MYC (binary)	BA 0.531	-0.084	Near chance or worse
MANTIS score (continuous)	Q2 0.107	-0.177	Below documented screening threshold
MSIsensor score (continuous)	Q2 0.094	-0.127	Below documented screening threshold
HALLMARK_ALLOGRAFT_REJECTION (continuous)	Q2 0.069	-0.163	Below documented screening threshold
HALLMARK_IL6_JAK_STAT3_SIGNALING (continuous)	Q2 0.134	-0.072	Below documented screening threshold
HALLMARK_INTERFERON_ALPHA_RESPONSE (continuous)	Q2 0.058	-0.156	Below documented screening threshold
HALLMARK_INTERFERON_GAMMA_RESPONSE (continuous)	Q2 0.177	-0.074	Below documented screening threshold
HALLMARK_MYOGENESIS (continuous)	Q2 0.152	-0.068	Below documented screening threshold
Macrophage Regulation (continuous)	Q2 -0.025	-0.251	Near chance or worse
Nonsilent Mutation Rate (continuous)	Q2 -0.006	-0.281	Near chance or worse
SNV Neoantigens (continuous)	Q2 0.015	-0.226	Below documented screening threshold
Silent Mutation Rate (continuous)	Q2 -0.004	-0.263	Near chance or worse

Case context (not used as model input)

Shared selection features. Both cases were men with sigmoid-colon, moderately differentiated (G2), pT3 adenocarcinoma and tumour-free resection margins. Nodal status was not matched: example A was pN1, whereas the available slide summary for example B did not state a nodal category.

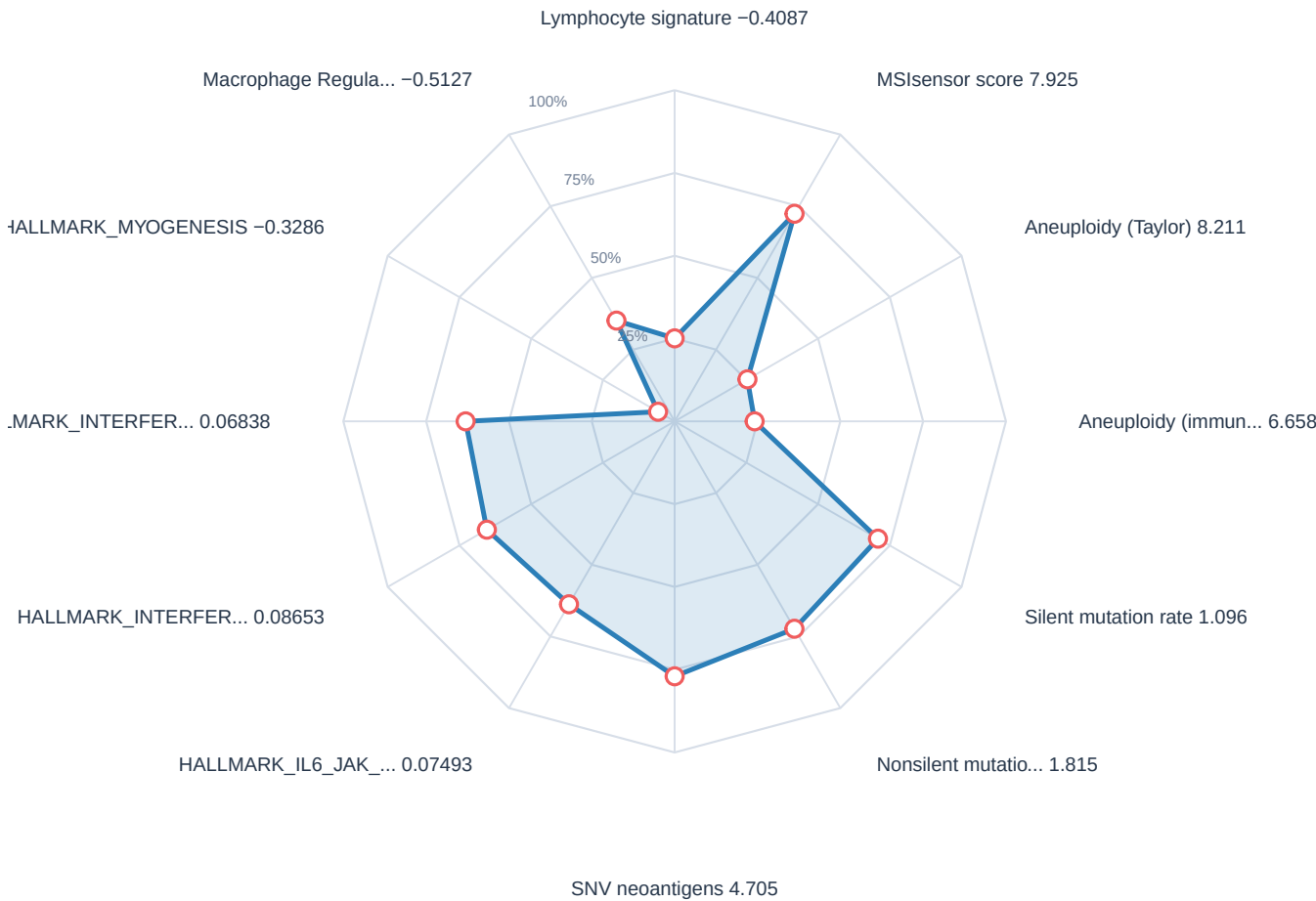
Pathology summary. The sigmoid resection contained an ulcerated grade-2 adenocarcinoma that had crossed the muscular wall into pericolic fat (pT3). Both resection ends were free of carcinoma; lymphatic invasion was recorded, and 2 of 30 regional nodes contained metastasis (pN1).

Clinical-context source. Pathology was paraphrased from TCGA-Slide-Reports.csv distributed with the TITAN study (Ding et al., Nature Medicine 2025; doi:10.1038/s41591-025-03982-3).

The pathology field is descriptive context only; it was not a predictor, training target, or validation outcome for PathoFMPred.

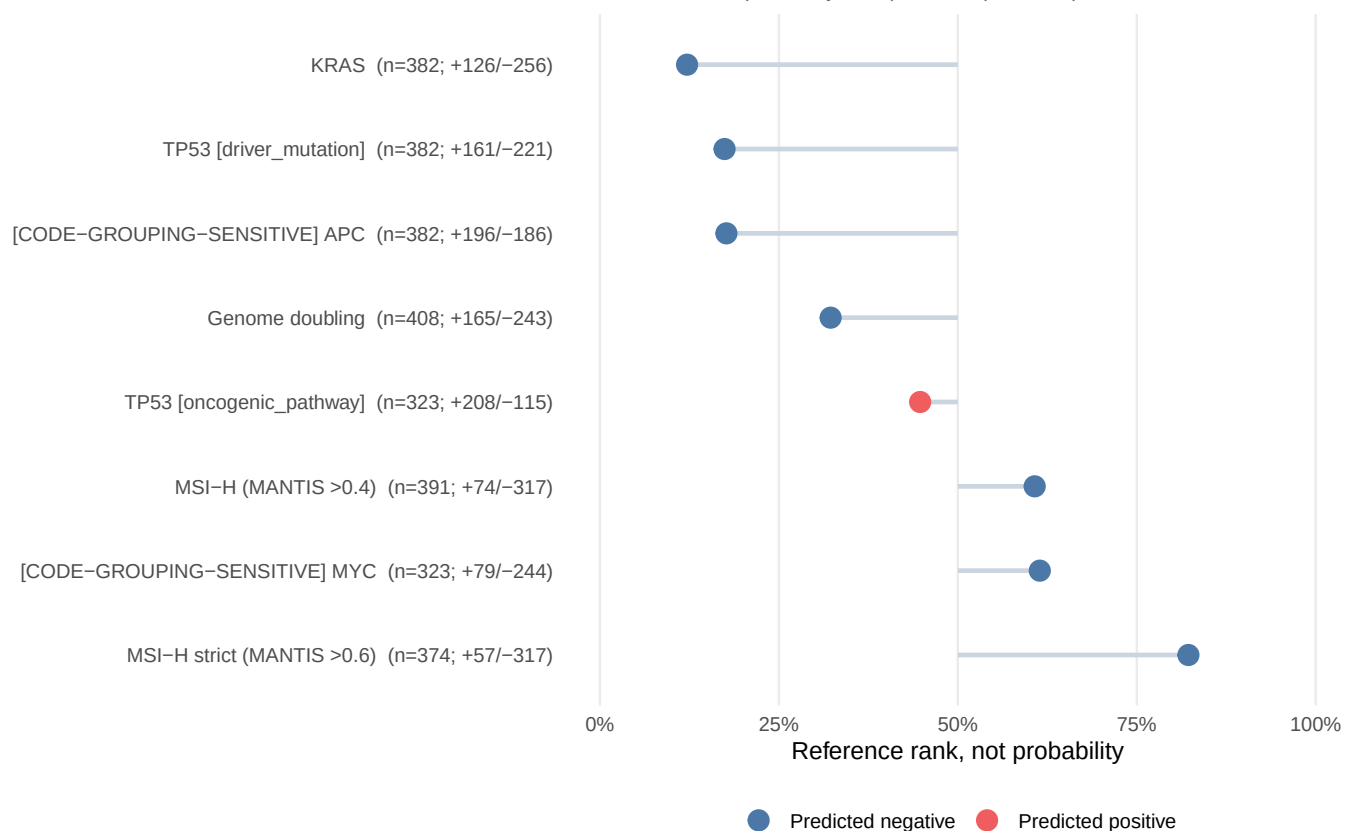
Continuous internally derived TCGA estimates

Only endpoints predictable for this cancer and representation are shown.



Binary research-model calls

Reference rank, not probability. Red points are predicted positive.



Continuous internally derived estimates

Family	Endpoint	Prediction	TCGA OOF rank (not probability)
rna_pathway_activity	HALLMARK_MYOGENESIS	-0.3286	5.8%
thorsson	SNV Neoantigens	4.705	77.0%
thorsson	Aneuploidy Score	6.658	24.1%
thorsson	Lymphocyte Infiltration Signature Score	-0.4087	25.1%
aneuploidy	Aneuploidy score	8.211	25.4%
microsatellite_instability	MSIsensor score	7.925	72.4%
thorsson	Nonsilent Mutation Rate	1.815	72.3%
thorsson	Silent Mutation Rate	1.096	70.9%
rna_pathway_activity	HALLMARK_INTERFERON_ALPHA_RESPONSE	0.08653	65.4%
thorsson	Macrophage Regulation	-0.5127	35.1%
rna_pathway_activity	HALLMARK_IL6_JAK_STAT3_SIGNALING	0.07493	63.8%
rna_pathway_activity	HALLMARK_INTERFERON_GAMMA_RESPONSE	0.06838	63.1%
microsatellite_instability	MANTIS score	0.4486	61.5%
thorsson	TIL Regional Fraction	9.861	56.4%
rna_pathway_activity	HALLMARK_ALLOGRAFT_REJECTION	-0.02422	46.8%

Internal development evidence. The primary screen and five-repeat estimates use different nested-CV partitions and need not be identical.

Endpoint	Primary screen Q2	Five-repeat Q2	Five-repeat RMSE	Five-repeat Spearman
Aneuploidy score	0.302	0.276	6.963	0.563
MANTIS score	0.284	0.229	0.203	0.401
MSIsensor score	0.222	0.247	9.529	0.269
HALLMARK_ALLOGRAFT_REJECTION	0.233	0.225	0.416	0.478
HALLMARK_IL6_JAK_STAT3_SIGNALING	0.206	0.181	0.399	0.447
HALLMARK_INTERFERON_ALPHA_RESPONSE	0.215	0.199	0.485	0.448
HALLMARK_INTERFERON_GAMMA_RESPONSE	0.251	0.239	0.442	0.499
HALLMARK_MYOGENESIS	0.220	0.187	0.336	0.415
Aneuploidy Score	0.217	0.266	5.959	0.556
Lymphocyte Infiltration Signature Score	0.297	0.299	0.719	0.547
Macrophage Regulation	0.226	0.196	0.567	0.446
Nonsilent Mutation Rate	0.275	0.207	0.826	0.309
SNV Neoantigens	0.241	0.215	0.964	0.353
Silent Mutation Rate	0.259	0.244	0.650	0.336
TIL Regional Fraction	0.407	0.404	6.533	0.689

Binary research-model calls

Calls and development denominators.

Endpoint	Class	LDA score	Threshold	Rank, not probability	Training n (+/-)
Genome doubling	0	-2.153	NA	32.2%	408 (+165/-243)
APC	0	-2.982	NA	17.7%	382 (+196/-186)
KRAS	0	-5.333	NA	12.2%	382 (+126/-256)
TP53 [driver_mutation]	0	-3.171	NA	17.4%	382 (+161/-221)
MSI-H (MANTIS >0.4)	0	-3.146	NA	60.7%	391 (+74/-317)
MSI-H strict (MANTIS >0.6)	0	-2.097	NA	82.2%	374 (+57/-317)
MYC	0	-1.27	NA	61.5%	323 (+79/-244)
TP53 [oncogenic_pathway]	1	0.8362	NA	44.7%	323 (+208/-115)

Internal development evidence. Primary AUROC and PR-AUC use nested held-out scores. The no-skill PR-AUC reference equals prevalence. Operating metrics and predictive values use training-derived thresholds.

Endpoint	AUROC	PR-AUC	Prevalence	No-skill PR-AUC	Threshold BA
Genome doubling	0.727	0.644	0.404	0.404	0.665
APC	0.834	0.822	0.513	0.513	0.777
KRAS	0.751	0.559	0.330	0.330	0.676
TP53 [driver_mutation]	0.822	0.789	0.421	0.421	0.767
MSI-H (MANTIS >0.4)	0.836	0.626	0.189	0.189	0.732
MSI-H strict (MANTIS >0.6)	0.877	0.714	0.152	0.152	0.783
MYC	0.662	0.375	0.245	0.245	0.616
TP53 [oncogenic_pathway]	0.735	0.815	0.644	0.644	0.693

Cohort-specific predictive values.

Endpoint	PPV	NPV	Sample-size stratum
Genome doubling	0.623	0.749	Larger-sample stratum
APC	0.759	0.851	Larger-sample stratum
KRAS	0.580	0.785	Larger-sample stratum
TP53 [driver_mutation]	0.710	0.824	Larger-sample stratum
MSI-H (MANTIS >0.4)	0.662	0.894	Larger-sample stratum
MSI-H strict (MANTIS >0.6)	0.710	0.929	Larger-sample stratum
MYC	0.451	0.794	Larger-sample stratum
TP53 [oncogenic_pathway]	0.755	0.638	Larger-sample stratum

Binary development reliability

Fold class counts and component selection.

Endpoint	Minimum outer test (+/-)	Minimum inner training (+/-)	Components: median (range)
Genome doubling	+33 / -48	+105 / -155	8.0 (2-16)
APC	+39 / -37	+124 / -118	2.0 (1-5)
KRAS	+25 / -51	+80 / -163	4.0 (2-20)
TP53 [driver_mutation]	+32 / -44	+102 / -140	2.0 (2-7)
MSI-H (MANTIS >0.4)	+14 / -63	+47 / -202	6.0 (4-13)
MSI-H strict (MANTIS >0.6)	+11 / -63	+36 / -202	8.0 (5-16)
MYC	+15 / -48	+50 / -156	4.0 (3-8)
TP53 [oncogenic_pathway]	+41 / -23	+132 / -73	3.0 (1-7)

Repeat-to-repeat held-out prediction stability.

Endpoint	Score Spearman rho	Class agreement
Genome doubling	0.760	0.829
APC	0.943	0.942
KRAS	0.732	0.837
TP53 [driver_mutation]	0.929	0.923
MSI-H (MANTIS >0.4)	0.801	0.919
MSI-H strict (MANTIS >0.6)	0.759	0.945
MYC	0.821	0.892
TP53 [oncogenic_pathway]	0.862	0.900

Quality and provenance

- Patient-level aggregation: arithmetic mean across all supplied slide vectors.
- Maximum fraction outside endpoint-specific TCGA training ranges: 0.3%.
- Foundation-model input schema: model-specific named features validated against the fitted artifact.
- Decomposition: seeded rSVD in fastPLS; binary classification: PLS-LDA.
- TCGA tissue-source-site-code metadata: every row includes the grouped internal metric, random-minus-grouped change, analysed-code count, threshold-retention status, fold-size range and inner-code-separation audit.
- Binary reliability metadata: held-out average precision, cohort-specific PPV/NPV, outer and inner class-count minima, selected-component distribution and repeat-to-repeat prediction stability.
- Default model selection: binary models require at least 50 TCGA participants in each class; smaller-class models are available only by explicit opt-in.
- Scope: internal validation grouped by TCGA tissue-source-site code is not institutional or scanner-level external validation.
- Intended use: research-software demonstration only; TCGA benchmark without independent external validation.

Endpoint interpretation and target sources

For continuous endpoints, the TCGA OOF prediction percentile (not probability) locates the prediction within repeated TCGA out-of-fold predictions and is not a clinical reference interval. For binary endpoints, every Reference rank, not probability value is the relative position of an uncalibrated LDA score. It is not confidence or a clinical-risk estimate. PR-AUC is non-interpolated average precision for positive class 1. PPV and NPV are held-out estimates at the observed endpoint prevalence in this TCGA analysis and will change when prevalence changes. Rows marked exploratory/limited evidence are excluded from default inference and require explicit opt-in.

- **Genome doubling.** Binary whole-genome-doubling call derived from the tumour copy-number profile. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* Taylor et al., Cancer Cell 2018, Table S2; doi:10.1016/j.ccell.2018.03.007.
- **APC.** Binary prediction of a qualifying protein-altering PASS mutation in APC; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* MC3 consensus calls: Ellrott et al., Cell Systems 2018, doi:10.1016/j.cels.2018.03.002; cancer-gene eligibility: Bailey et al., Cell 2018, doi:10.1016/j.cell.2018.02.060.
- **KRAS.** Binary prediction of a qualifying protein-altering PASS mutation in KRAS; the displayed LDA score is not a probability. *Analysed scale:* PLS-LDA class and uncalibrated discriminant score *Model-target source:* MC3

consensus calls: Ellrott et al., Cell Systems 2018, doi:10.1016/j.cels.2018.03.002; cancer-gene eligibility: Bailey et al., Cell 2018, doi:10.1016/j.cell.2018.02.060.

- **TP53 [driver_mutation]**. Binary prediction of a qualifying protein-altering PASS mutation in TP53; the displayed LDA score is not a probability. *Analysed scale*: PLS-LDA class and uncalibrated discriminant score *Model-target source*: MC3 consensus calls: Ellrott et al., Cell Systems 2018, doi:10.1016/j.cels.2018.03.002; cancer-gene eligibility: Bailey et al., Cell 2018, doi:10.1016/j.cell.2018.02.060.
- **MSI-H (MANTIS >0.4)**. Binary call indicating whether the source MANTIS score exceeds 0.4; the displayed LDA score is not a probability. *Analysed scale*: PLS-LDA class and uncalibrated discriminant score *Model-target source*: Bonneville et al., JCO Precision Oncology 2017; doi:10.1200/PO.17.00073.
- **MSI-H strict (MANTIS >0.6)**. Sensitivity-analysis call: MANTIS >0.6 is positive, <0.4 is negative, and intermediate source values were excluded from training. *Analysed scale*: PLS-LDA class and uncalibrated discriminant score *Model-target source*: Threshold documentation distributed with the cBioPortal TCGA PanCancer Atlas clinical sample files.
- **MYC**. Binary prediction that the MYC oncogenic pathway carries at least one qualifying genomic alteration; the displayed LDA score is not a probability. *Analysed scale*: PLS-LDA class and uncalibrated discriminant score *Model-target source*: Sanchez-Vega et al., Cell 2018, Table S4; doi:10.1016/j.cell.2018.03.035.
- **TP53 [oncogenic_pathway]**. Binary prediction that the TP53 oncogenic pathway carries at least one qualifying genomic alteration; the displayed LDA score is not a probability. *Analysed scale*: PLS-LDA class and uncalibrated discriminant score *Model-target source*: Sanchez-Vega et al., Cell 2018, Table S4; doi:10.1016/j.cell.2018.03.035.
- **Aneuploidy score**. Number of chromosome arms with arm-level copy-number gain or loss; larger values indicate greater chromosomal imbalance. *Analysed scale*: source endpoint units *Model-target source*: Taylor et al., Cancer Cell 2018, Table S2; doi:10.1016/j.ccell.2018.03.007.
- **MANTIS score**. Genome-wide microsatellite-instability score from the MANTIS algorithm; larger values indicate stronger MSI evidence. *Analysed scale*: source endpoint units *Model-target source*: Kautto et al., Oncotarget 2017; doi:10.18632/oncotarget.20432. TCGA values were obtained from cBioPortal PanCancer Atlas records.
- **MSIsensor score**. Microsatellite-instability score from MSIsensor; larger values indicate a larger fraction of unstable microsatellite loci. *Analysed scale*: source endpoint units *Model-target source*: Niu et al., Bioinformatics 2014; doi:10.1093/bioinformatics/btt755. TCGA values were obtained from cBioPortal PanCancer Atlas records.
- **HALLMARK_ALLOGRAFT_REJECTION**. Model output for the named endpoint. Interpret together with its family, analysed scale and source publication. *Analysed scale*: source endpoint units *Model-target source*: See the PathoFMPred model registry and the companion analysis source manifest.
- **HALLMARK_IL6_JAK_STAT3_SIGNALING**. Model output for the named endpoint. Interpret together with its family, analysed scale and source publication. *Analysed scale*: source endpoint units *Model-target source*: See the PathoFMPred model registry and the companion analysis source manifest.
- **HALLMARK_INTERFERON_ALPHA_RESPONSE**. Model output for the named endpoint. Interpret together with its family, analysed scale and source publication. *Analysed scale*: source endpoint units *Model-target source*: See the PathoFMPred model registry and the companion analysis source manifest.
- **HALLMARK_INTERFERON_GAMMA_RESPONSE**. Model output for the named endpoint. Interpret together with its family, analysed scale and source publication. *Analysed scale*: source endpoint units *Model-target source*: See the PathoFMPred model registry and the companion analysis source manifest.
- **HALLMARK_MYOGENESIS**. Model output for the named endpoint. Interpret together with its family, analysed scale and source publication. *Analysed scale*: source endpoint units *Model-target source*: See the PathoFMPred model registry and the companion analysis source manifest.
- **Aneuploidy Score**. Aneuploidy burden distributed with the TCGA immune landscape resource; larger values indicate more arm-level copy-number alterations. *Analysed scale*: source endpoint units *Model-target source*: Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.
- **Lymphocyte Infiltration Signature Score**. Gene-expression signature summarising lymphocyte-associated transcriptional infiltration; it is not a direct cell count. *Analysed scale*: source endpoint units *Model-target source*: Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.

- **Macrophage Regulation.** Model output for the named endpoint. Interpret together with its family, analysed scale and source publication. *Analysed scale:* source endpoint units *Model-target source:* See the PathoFMPred model registry and the companion analysis source manifest.
- **Nonsilent Mutation Rate.** Rate of coding mutations that alter the encoded protein. PathoFMPred reports the model output on the analysis $\log(1+x)$ scale. *Analysed scale:* $\log_{1p}$ -transformed source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.
- **SNV Neoantigens.** Predicted neoantigen burden arising from single-nucleotide variants. PathoFMPred reports the model output on the analysis $\log(1+x)$ scale. *Analysed scale:* $\log_{1p}$ -transformed source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.
- **Silent Mutation Rate.** Rate of coding mutations that do not change the encoded amino acid. PathoFMPred reports the model output on the analysis $\log(1+x)$ scale. *Analysed scale:* $\log_{1p}$ -transformed source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.
- **TIL Regional Fraction.** Estimated fraction of tumour image regions containing tumour-infiltrating lymphocytes in the TCGA immune landscape resource. *Analysed scale:* source endpoint units *Model-target source:* Thorsson et al., Immunity 2018, PanImmune_MS; doi:10.1016/j.immuni.2018.03.023.